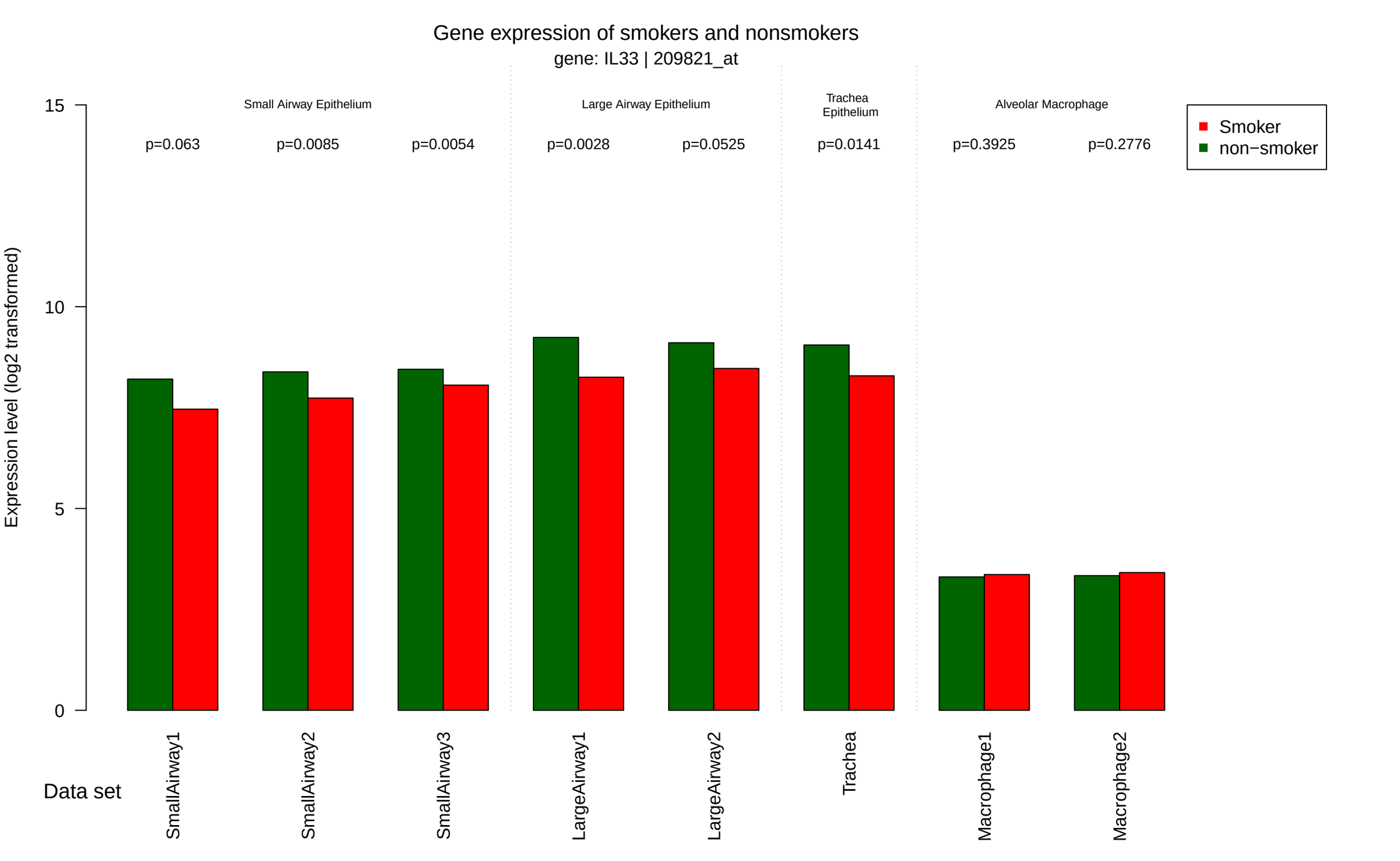
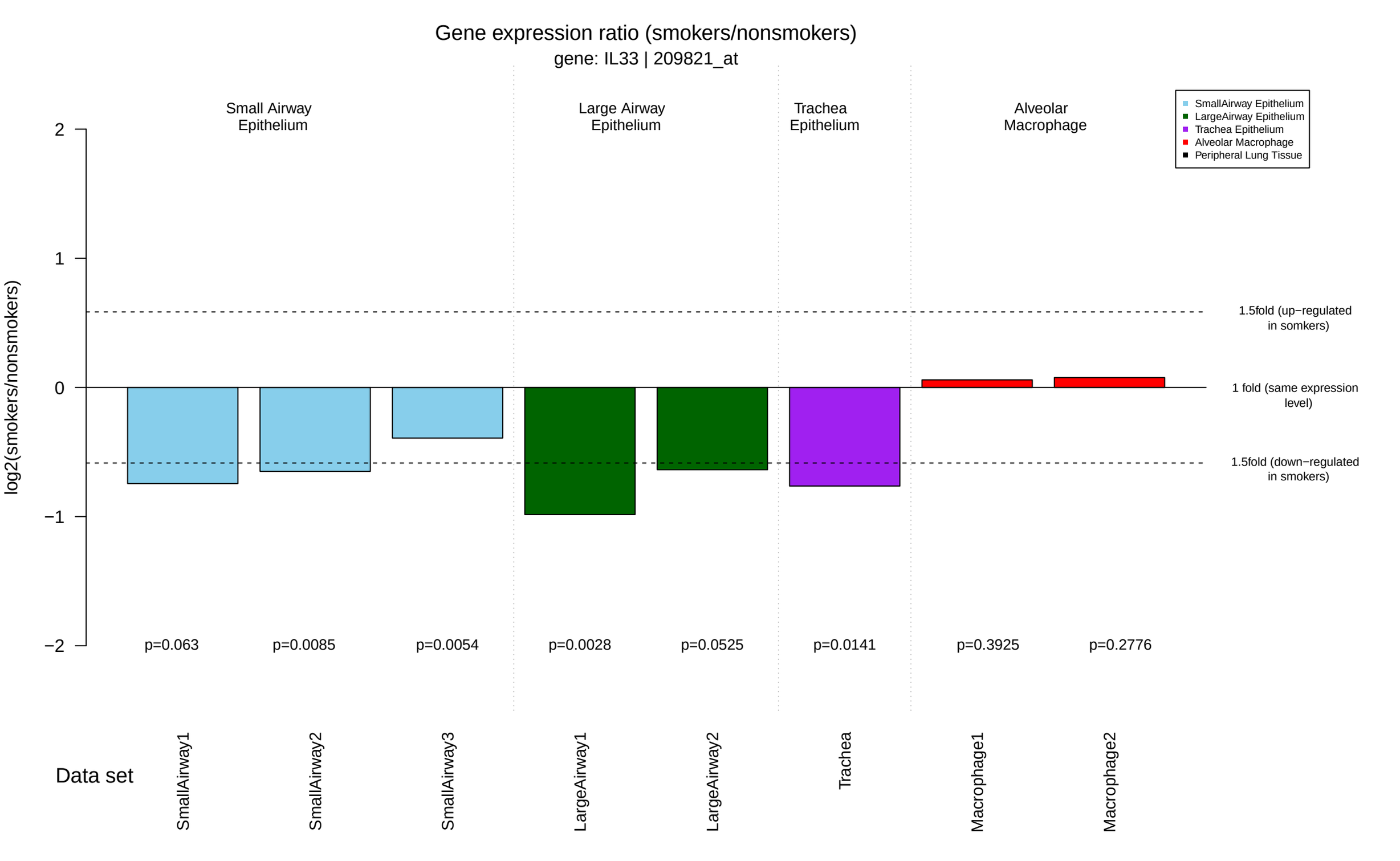
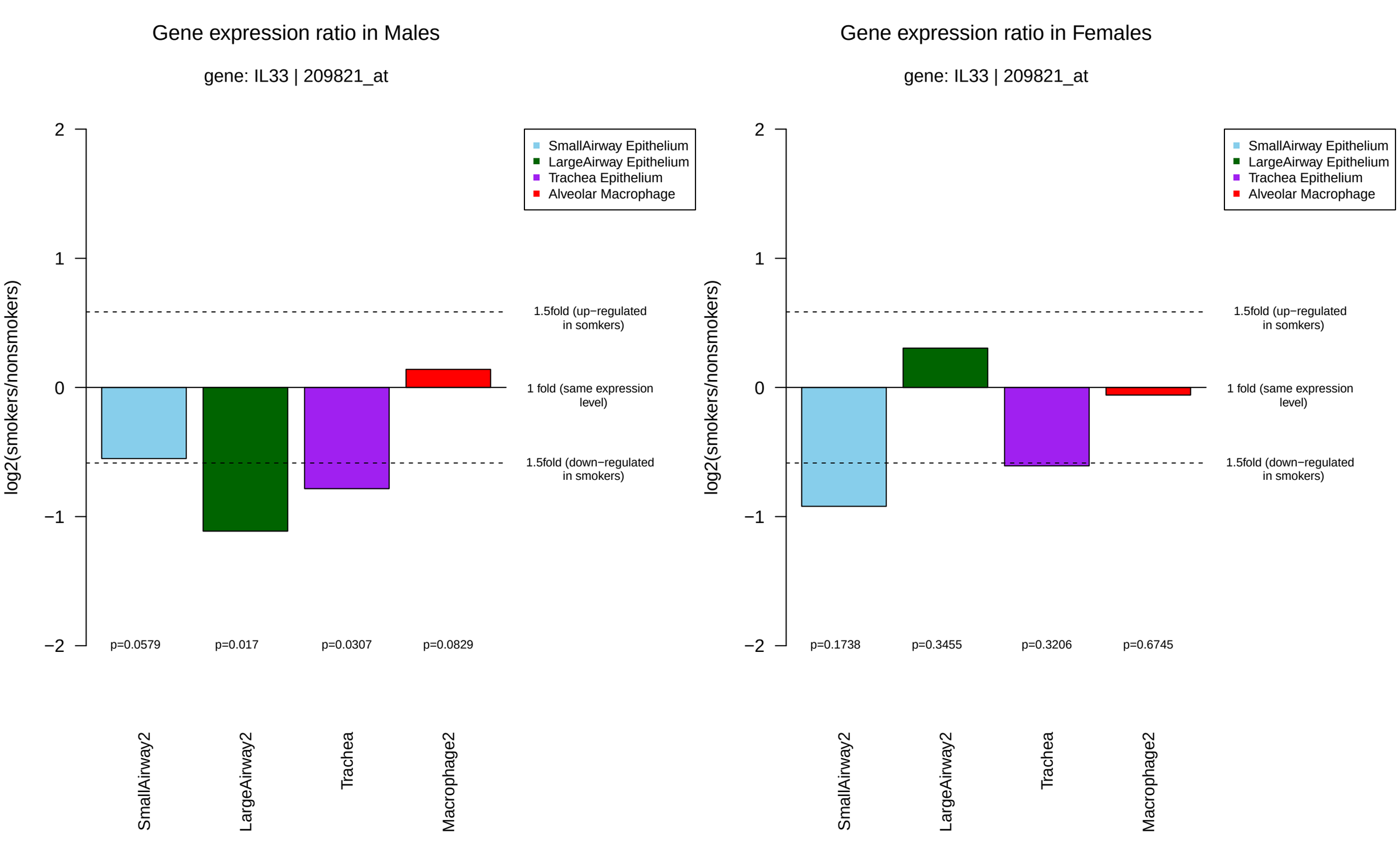
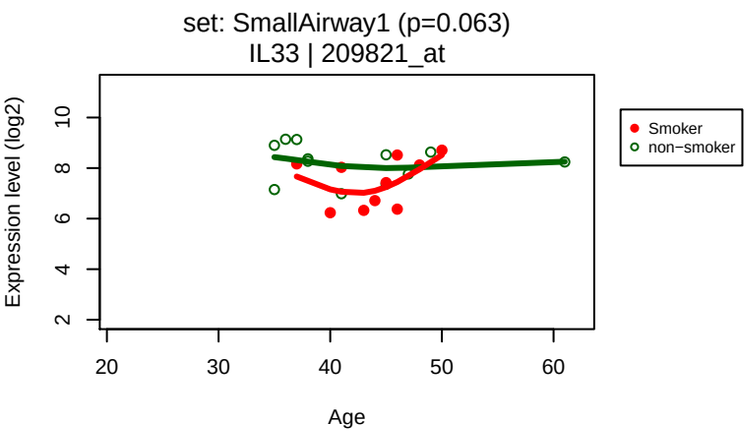
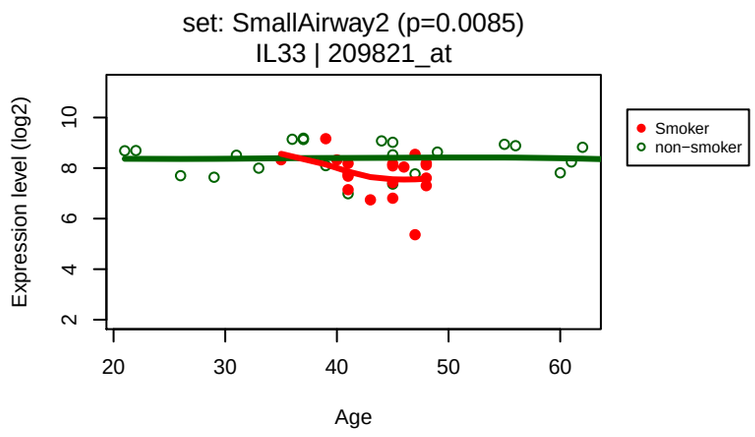
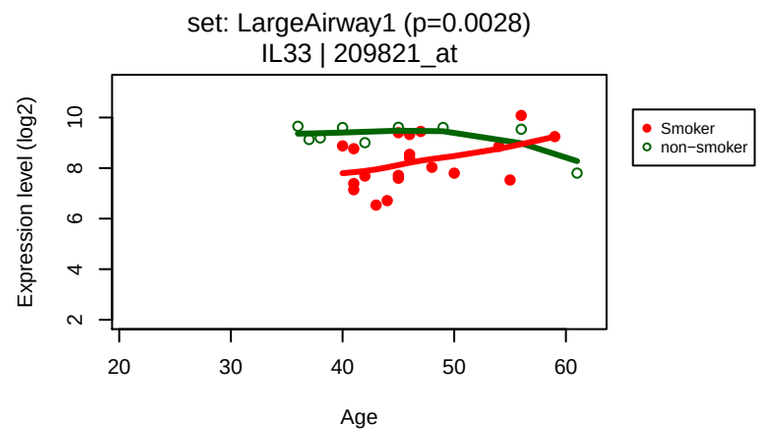

Gene expression of smokers and nonsmokers
gene: IL33 | 209821_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.063
p=0.0085
p=0.0054
p=0.0028
p=0.0525
p=0.0141
p=0.3925
p=0.2776
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

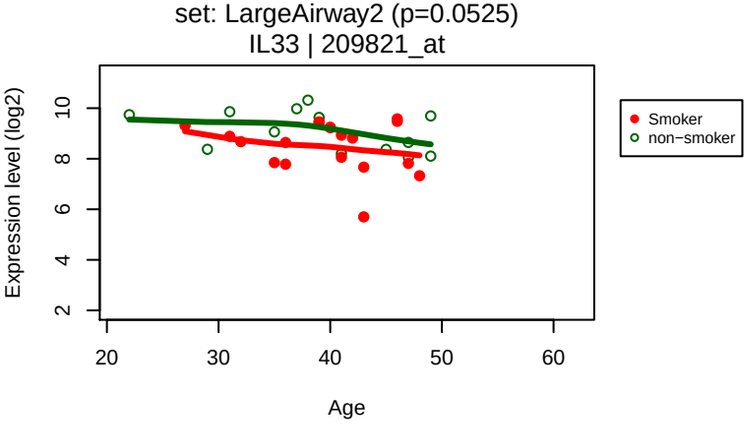

Gene expression ratio (smokers/nonsmokers)
gene: IL33 | 209821_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.063
p=0.0085
p=0.0054
p=0.0028
p=0.0525
p=0.0141
p=0.3925
p=0.2776
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

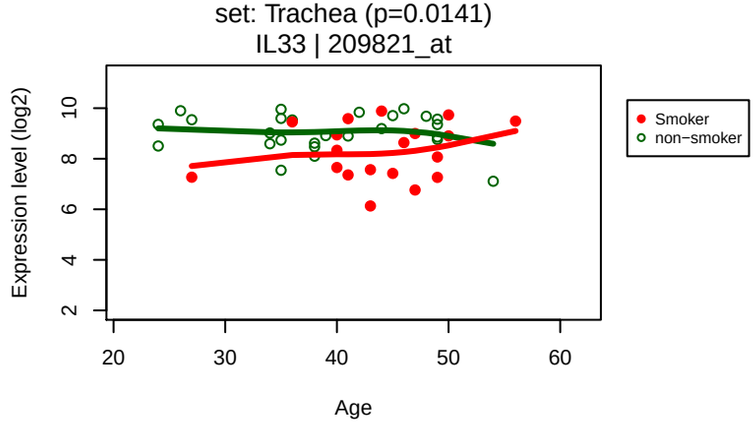

Gene expression ratio in Males
gene: IL33 | 209821_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0579
p=0.017
p=0.0307
p=0.0829
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IL33 | 209821_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1738
p=0.3455
p=0.3206
p=0.6745
SmallAirway2
LargeAirway2
Trachea
Macrophage2

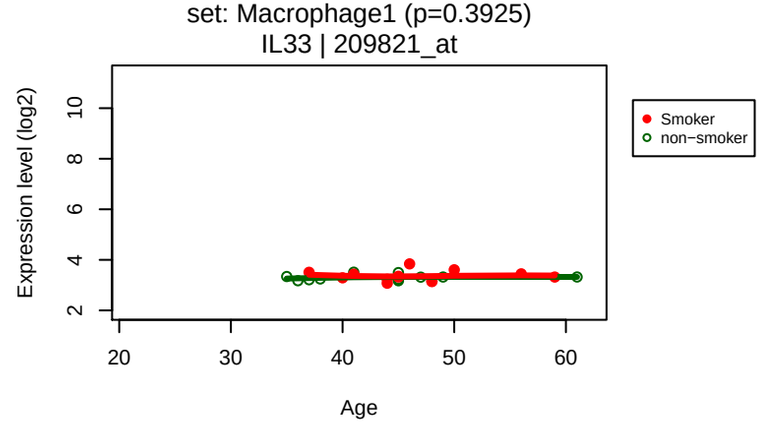

set: SmallAirway1 (p=0.063)
IL33 | 209821_at
Expression level (log2)
Age
Smoker
non−smoker

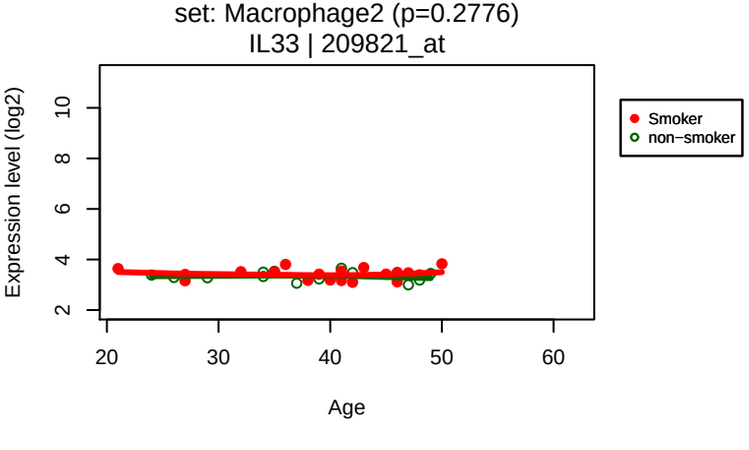

set: SmallAirway2 (p=0.0085)
IL33 | 209821_at
Expression level (log2)
Age
Smoker
non−smoker

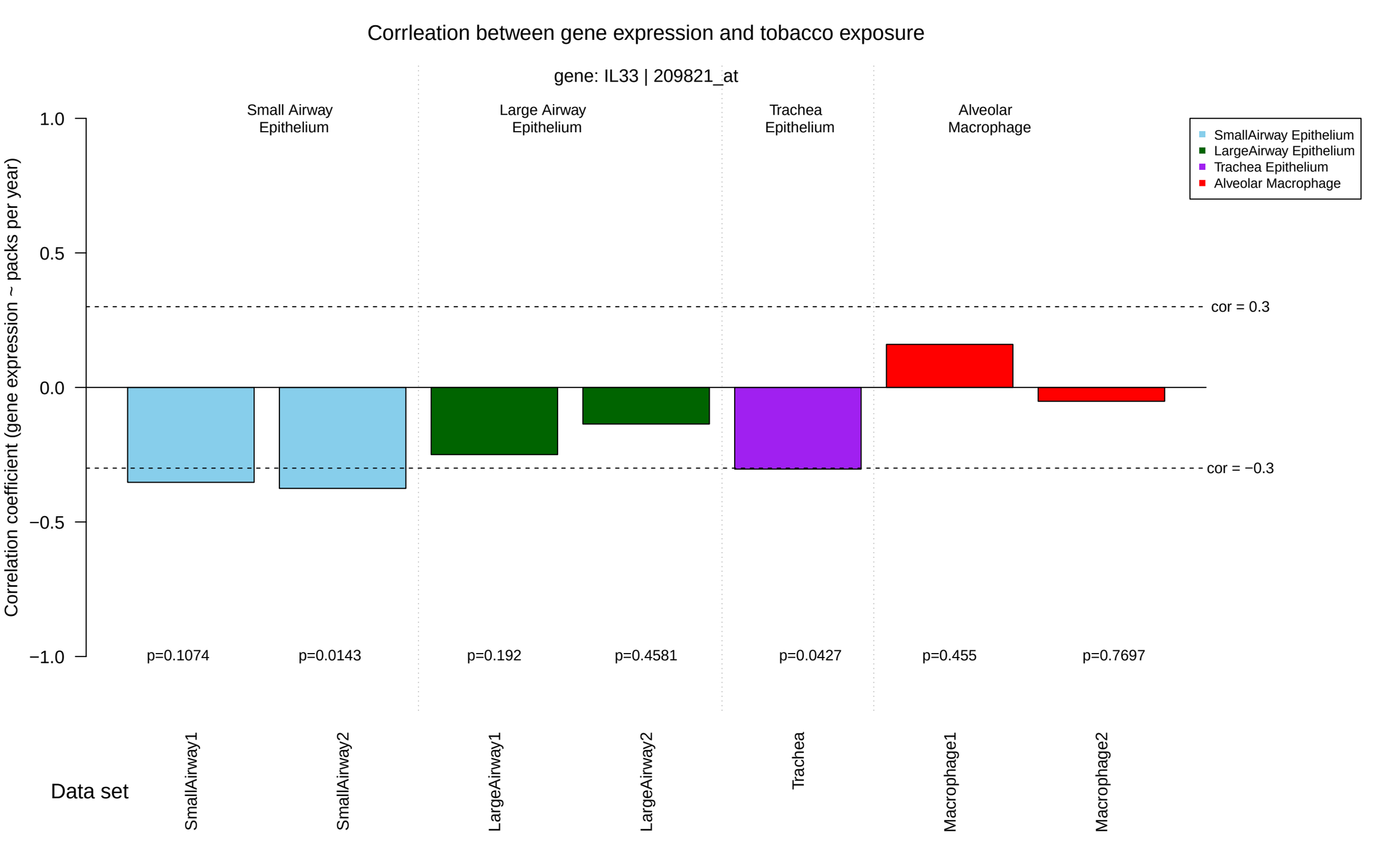

set: LargeAirway1 (p=0.0028)
IL33 | 209821_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0525)
IL33 | 209821_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0141)
IL33 | 209821_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3925)
IL33 | 209821_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2776)
IL33 | 209821_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IL33 | 209821_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1074
p=0.0143
p=0.192
p=0.4581
p=0.0427
p=0.455
p=0.7697
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2